L22

PDF

Role: You are an academic scribe for a University-level Probability course. Task: Create a comprehensive lecture scribe based exclusively on the attached lecture slides (Lecture 22:).Requirements :Structure: Organize the content logically with sections matching the slide topic...

L22.pdf

Analysis

This lecture scribe covers **Lecture 22** of the **CSE400 Probability** course, focusing on **Markov's** and **Chebyshev's Inequalities**. These tools provide probabilistic bounds on how much a random variable can deviate from its mean. +2

1. Markov's Inequality

L22.pdf

Markov's Inequality provides an upper bound on the probability that a non-negative random variable X is greater than or equal to a positive constant a.

Definition

For a non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

+1

2. Chebyshev's Inequality

Chebyshev's Inequality uses both the mean (μ) and the variance (σ^2) to contr L22.pdf viation of a random variable from its mean.

Theorem

For a random variable X with mean μ and variance $Var(X)$:

$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

+1

Proof Idea

The proof is derived by applying **Markov's Inequality** to the squared deviation $(X - \mu)^2$:

1. Express the probability of the absolute deviation as a squared deviation: L22.pdf

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$

2. Apply Markov's Inequality:

$$P((X - \mu)^2 \geq a^2) \leq \frac{E[(X - \mu)^2]}{a^2}$$

3. Substitute the definition of variance ($Var(X) = E[(X - \mu)^2]$):

$$\leq \frac{Var(X)}{a^2}$$

L22.pdf

Intuition

The inequality bounds the "tail regions" of a distribution—the areas far from the mean μ (specifically, values $\leq \mu - a$ or $\geq \mu + a$). +1

3. Comparative Example: Binomial Distribution

Consider a fair coin flipped n times ($X \sim \text{Binomial}(n, 0.5)$): +1

- **Parameters:** $\mu = \frac{n}{2}$, $Var(X) = \frac{n}{4}$.
- **Problem:** Bound the probability that $X \geq \frac{3n}{4}$.

L22.pdf

- **Chebyshev Bound:**

- **Chebyshev Bound:**

$$P\left(X \geq \frac{3}{4}n\right) = \frac{1}{2}P\left(|X - \frac{n}{2}| \geq \frac{n}{4}\right) \leq \frac{2}{n}$$

- **Key Insight:** Unlike Markov's Inequality, which provides a constant bound for this setup, the Chebyshev bound **decreases as n increases**, providing a tighter and more useful estimate for large n.

---

4. Solved Examples

L22.pdf

Example 1: Exam Scores (General Bound)

Scenario: Exam scores have a mean (μ) of 70 and a standard deviation (σ) of 10.
Question: What fraction of students score below 40 or above 100?
Solution: +1

1. Identify the deviation: $a = 30$ (since $|40 - 70| = 30$ and $|100 - 70| = 30$).
2. Apply Chebyshev:

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$$

Result: At most $\frac{1}{9}$ (approx. 11.1%) of students are far from the mean. +1

L22.pdf

Example 2: Exam Grades (One-Sided Application)

Scenario: Average grade is 70%, $\sigma = 5\%$. The "A" cutoff is 90%.
Problem: Find a tighter upper bound on the fraction of "A" students.
Solution: +1

1. **Parameters:** $\mu = 70, \sigma^2 = 25$.
2. **Relate to Absolute Deviation:** $X \geq 90 \implies |X - 70| \geq 20$. Therefore, $P(X \geq 90) \leq P(|X - 70| \geq 20)$.
3. **Apply Chebyshev ($a = 20$):**

$$P(|X - 70| \geq 20) \leq \frac{25}{20^2} = \frac{25}{400} = \frac{1}{16}$$

Result: $P(X \geq 90) \leq 0.0625$. This is significantly tighter than the Markov bound of 0.778.

L22.pdf

Example 3: Network Latency SLA

Scenario: Average latency $E[X] = 50$ ms, $Var(X) = 16$ ms^2.
Problem: Bound the probability that latency falls outside the range $[30 \text{ ms}, 70 \text{ ms}]$.
Solution:

1. **Parameters:** $\mu = 50, \sigma^2 = 16$.
2. **Condition:** Falling outside $[30, 70]$ is equivalent to $|X - 50| \geq 20$.
3. **Apply Chebyshev ($a = 20$):**

$$P(|X - 50| \geq 20) \leq \frac{16}{20^2} = \frac{16}{400} = \frac{1}{25}$$

Result: At most 4% of packets fall outside the specified range.

L22.pdf

---

5. Homework Problem: Manufacturing Tolerances

L22.pdf

Problem: A factory produces resistors with $\mu = 100\Omega$ and $\sigma = 2\Omega$. A resistor is defective if it lies outside $[94\Omega, 106\Omega]$.

- **(a) Upper bound on defective probability:** Express "defective" as $|X - \mu| \geq a$. Here $a = 6$ and $\sigma^2 = 4$. Apply $P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$.
- **(b) Maximum expected defective units (for 10,000 produced):** Multiply the probabilistic bound from part (a) by $10,000$.

Sources

Enter a prompt for Gemini

Tools

Fast

Your Ahmedabad University chats aren't used to improve our models. Gemini is AI and can make mistakes. Your privacy & Gemini